Passage-7

The nature of the legal imperatives in any given state corresponds to the effective demands that state encounters. And that these, in their turn, depend, in a general way, upon the manner in which economic power is distributed in the society which the state controls.

47. The statement refers to :

(a) the antithesis of Politics and Economics
(b) the interrelationship of Politics and Economics
(c) the predominance of Economics over Politics
(d) The predominance of Politics over Economics

Ans. (b) : The passage refers to the interrelationships between politics and economy. According to the demands of the majority of the people, laws are made in which economic interests are embedded.

Passage -8

About 15 percent of global greenhouse gas emissions come from agricultural practices. This includes nitrous oxide from fertilizers; methane from livestock, rice production, and manure storage; and carbon dioxide (CO_2) from burning biomass, but this excludes CO_2 emissions from soil management practices, savannah burning and deforestation Forestry land use, and land-use change account for another 17 percent of greenhouse gas emissions each year, three quarters of which come from tropical deforestation. The remainder is largely from draining and burning tropical peat land. About the same amount of carbon is stored in the world's peat lads as is stored in the Amazon rainforest.

48. Which among the following is the most logical and rational inference that can be made from the above passage?

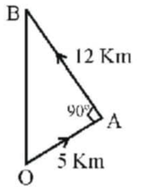
(a) Organic farming should immediately replace mechanised and chemical dependant agricultural practices all over the world
(b) It is imperative for us to modify our land use practices in order to mitigate climate change.
(c) There are no technological solutions to the problem of greenhouse gas emissions
(d) Tropical areas are the chief sites of carbon sequestration.

Ans. (b) : The most logical and rational inference that can be drawn from the above passage is that, it is imperative for us to modify our land use practices in order to mitigate climate change.

49. A person climbs a hill in a straight path from point 'O' on the ground in the direction of north east and reaches a point 'A' after traveling a distance of 5 km. Then, from the point 'A' he moves to point 'B' in the direction of north-west. Let the distance AB be 12 km. Now, how far is the person away from the starting point 'O'?

(a) 7 km (b) 13 km
(c) 17 km (d) 11 km

Ans. (b) :



Distance from point B to starting point 'O'
By Pythagoras theorem-

$= \sqrt{5^2 + 12^2}$
$= \sqrt{25 + 144}$
$= \sqrt{169} = 13$ Km.

50. An agricultural field is in the from of a rectangle having length X_1 meters and breadth X_2 meters (X_1 and X_2 are variable) . If X_1+X_2 = 40 meters then the area of the agricultural field will not exceed which one of the following values ?

(a) 400 sq m (b) 300 sq m
(c) 200 sq m (d) 80 sq m

Ans. (a) :
Given, length = X_1 m and breadth = X_2 m Also, $X_1 + X_2 = 40$, where X_1 and X_2 are variables we know that, of all the rectangles, a square has the largest area.
For the given rectangle to be a square $X_1 = X_2$
So, $X_1 + X_2 = 40 \Rightarrow X_1 = X_2 = 20$ m
Maximum area of the given rectangle
$= X_1X_2 = (20 \times 20) = 400\ m^2$

51. The sum of the ages of 5 members comprising a family, 3 years ago was 80 years. The average age of the family today is the same as it was 3 year age, because of an addition of a baby during the intervening period. How old is the baby ?

(a) 6 months
(b) 1 years
(c) 2 year
(d) 2 year and 6 month

Ans. (b) : The average age of family 3 years ago was

$= \frac{80}{5} = 16$ years

The sum of the ages of 5 members today will be = 80 + (3 × 5) = 80 + 15 = 95 years
Now, the average age of family including new body will be = $\frac{95+N}{6}$, which is same as 3 years ago, i.e., 16 years.
N is the age of baby now

$\therefore \frac{95+N}{6} = 16 \Rightarrow 95 + N = 16 \times 6$

N = 96 – 95 $\Rightarrow$ N = 1 years

52. The total emoluments of two persons are the same, but one gets allowances to the extent of 65% of his basic pay and the other gets allowances to the extent of 80% of his basic pay.The ratio of the basic pay of the former to the basic pay of the latter is ?

(a) 16 : 13 (b) 5 : 4
(c) 7 : 5 (d) 12 : 11

Ans. (d) :
Let the basic pay of the first person be 'x' and that of second person be 'y'.
According to the question,
Emolument of both the persons are same.
So, x + 0.65 x = y + 0.8y

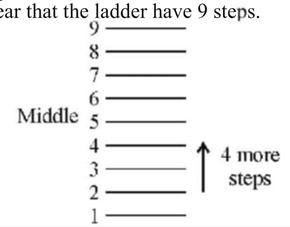
$\Rightarrow x(1+0.65) = y(1+0.8) \Rightarrow \frac{x}{y} = \frac{1.8}{1.65}$

$\Rightarrow \frac{x}{y} = \frac{12}{11}$ or 12 : 11

53. A person is standing on the first step from the bottom of a ladder . If he has to climb 4 more steps to reach exactly the middle step, how many steps does the ladder have ?

(a) 8 (b) 9
(c) 10 (d) 11

Ans. (b) :
A person is standing on the first step from the bottom of a ladder.
Now, he has to climb 4 more steps to reach exactly the middle step.
So, it is clear that the ladder have 9 steps.



Direction for the following 3 (three) items : Consider the given information and answer the three items that follow.

When three friends A, B and C met, it was found that each of them wore an outer garment of a different colour. In random order, the garments are: jacket, sweater and tie; and the colours are : bule , white and black . Their surnames in random order are: Ribeiro, Kumar, Singh we know that :

1. neither B nor Ribeiro wore a white sweater
2. C wore a tie
3. Singh's garment was not white
4. Kumar does not wear a jacket
5. Ribeiro does not like to wear the black colour
6. Each of the friends wore only one outer garment of only one colour

Solution for 54 to 56-

Person	A	B	C
Surnames	Kumar	Singh	Ribeiro
Garments	Sweater	Jacket	Tie
Colour	White	Black	Blue

54. What is C's surname ?

(a) Ribeiro
(b) Kumar
(c) Singh
(d) Cannot be determined

Ans. (a) : C's surname is Riberio.

55. What is the colour of tie ?

(a) Black
(b) Blue
(c) White
(d) Cannot be determined

Ans. (b) : The colour of tie is blue.

56. Who wore the sweater ?

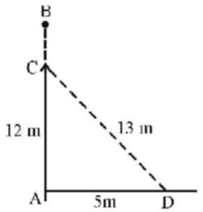
(a) A
(b) B
(c) C
(d) Cannot be determined

Ans. (a) : A wore the sweater.

57. A B is a vertical trunk of a huge tree with A being the point where the base of the trunk touches the ground. Due to a cyclone, the trunk has been broken at C which is at a height of 12 meters, broken part is partially attached to the vertical portion of the trunk at C. If the end of the broken part B touches the ground at D which is at a distance of 5 meters from A, then the original height of the trunk is:

(a) 20 m
(b) 25 m
(c) 30 m
(d) 35 m

Ans. (b) :



The distance between broken part of tree on surface and trunk of the tree = 5cm
Hypotenuses^2 = Perpendicular^2 + base^2
In, Δ CAD,

$CD^2 = AC^2 + AD^2$
$CD = \sqrt{AC^2 + AD^2}$
$= \sqrt{12^2 + 5^2} = \sqrt{144 + 25}$
$CD = \sqrt{169} = 13$ meter

AB = CA + CD
= (12 + 13) meter = 25 meter